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Power transmission device

Abstract

A power transmission device includes: a drive source; a transmission portion configured to convert power of the drive source and transmit the power to a driven portion; and a case accommodating the drive source and the transmission portion, in which the power transmission device further comprises a drive sprocket disposed at an output shaft of the drive source, a driven sprocket disposed on the same plane as the drive sprocket, an endless flexible member wound around the drive sprocket and the driven sprocket, and a parking gear disposed at the output shaft of the drive source.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6106428	12/1999	Koneda	475/210	F16H 37/022
11192436	12/2020	Puiu	N/A	B60K 17/16
2005/0205384	12/2004	Reed	192/219.5	F16H 57/031
2006/0070839	12/2005	Sugano	192/219.5	F16H 63/3416
2012/0145512	12/2011	Kim	192/219.5	F16H 63/3425
2022/0205534	12/2021	Morise	N/A	F16H 63/3466
2023/0204102	12/2022	Matsuo	192/219.5	F16H 63/3425

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H10-148241	12/1997	JP	N/A
20220169297	12/2021	KR	F16H 63/38

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2023-128807 filed on Aug. 7, 2023, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to a power transmission device.

BACKGROUND ART

(3) In related art, there has been known a power transmission device including a transmission portion in which an endless flexible member such as a chain or a belt is wound around a drive sprocket and a driven sprocket, and power is transmitted via the endless flexible member. In such a power transmission device, it is necessary to set a parking mechanism so as to avoid a trajectory of the endless flexible member.

(4) For example, in a remote video pan-tilt mechanism disclosed in JPH10-148241A, a mechanism is disposed inside a chain belt. The mechanism includes a first gear constantly meshed with a sprocket and a second gear constantly meshed with the first gear, and the second gear is meshed

with the sprocket while the first gear and the second gear are meshed with each other to interlock and stop drive.

(5) For example, a power transmission device used in an electric vehicle or the like is desired to be reduced in size in order to ensure as much space in a passenger compartment as possible.

Therefore, a degree of freedom in arrangement of a parking mechanism is required.

(6) An object of the present invention is to provide a power transmission device with a high degree of freedom in layout.

SUMMARY OF INVENTION

(7) According to an aspect of the present invention, there is provided a power transmission device including: a drive source; a transmission portion configured to convert power of the drive source and transmit the power to a driven portion; and a case accommodating the drive source and the transmission portion, in which the power transmission device further includes a drive sprocket disposed at an output shaft of the drive source, a driven sprocket disposed on the same plane as the drive sprocket, an endless flexible member wound around the drive sprocket and the driven sprocket, and a parking gear disposed at the output shaft of the drive source.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic view of a vehicle on which a drive unit **10** according to an embodiment of the present invention is mounted as viewed from a left side.

(2) FIG. 2 is a cutaway perspective view of a part of the drive unit **10** of the embodiment of the present invention.

(3) FIG. 3 is a partial cross-sectional perspective view showing a power transmission path in the drive unit **10** in FIG. 2.

(4) FIG. 4 is a skeleton diagram of a deceleration device in the drive unit **10** in FIG. 2.

(5) FIG. 5 is a perspective view showing an internal structure of a chain chamber **14**.

(6) FIG. 6 is a side view schematically showing a part of a chain transmission mechanism **4** and a parking mechanism **6**.

(7) FIG. 7 is a plan view schematically showing a part of the chain transmission mechanism **4** and the parking mechanism **6**.

(8) FIG. 8 is a schematic diagram of the chain transmission mechanism **4** showing an example of an arrangement condition of a control shaft **65**.

(9) FIG. 9 is a schematic diagram of the chain transmission mechanism **4** showing another example of the arrangement condition of the control shaft **65**.

(10) FIG. 10 is an explanatory diagram of a calculation formula for calculating a radius of an arc **L3** shown in FIG. 9.

DESCRIPTION OF EMBODIMENTS

(11) Hereinafter, a vehicle on which a drive unit according to an embodiment of the present invention will be described with reference to the accompanying drawings. The drawings are viewed in directions of reference numerals. In the present specification and the like, in order to simplify and clarify the description, a front-rear direction, a left-right direction, and an upper-lower direction are described according to directions viewed from a driver of the vehicle, and in the drawings, a front side of the vehicle is shown as Fr, a rear side is shown as Rr, a left side is shown as L, a right side is shown as R, an upper side is shown as U, and a lower side is shown as D.

(12) [Overall Configuration of Vehicle]

(13) As shown in FIG. 1, a vehicle V according to the present embodiment includes a pair of left and right front wheels FW, a pair of left and right rear wheels RW, and a floor panel FP constituting a floor of the vehicle V. The vehicle V is partitioned, by a dash panel DP extending in the upper-

lower direction above the floor panel FP, into a passenger compartment CB and a front room FRM in front of the passenger compartment CB. A front seat FS and a rear seat RS are provided in the passenger compartment CB.

(14) The vehicle V includes a drive unit **10** serving as a drive source, and a battery pack IPU that stores electric power to be supplied to the drive unit **10**.

(15) The drive unit **10** is disposed behind the rear seat RS and below the floor panel FP. The battery pack IPU is disposed below the floor panel FP and below a floor of the passenger compartment CB. The battery pack IPU accommodates a plurality of battery modules in which a plurality of battery cells are stacked. Each battery cell is a secondary battery that can be charged and discharged, such as a lithium ion battery or an all-solid-state battery.

(16) [Overall Configuration of Drive Unit]

(17) As shown in FIGS. 2 and 3, the drive unit **10** includes a drive motor **20**, a deceleration device **30** that reduces power output from the drive motor **20** and outputs the reduced power to the outside, a chain transmission mechanism **4** that transmits the power output from the drive motor **20** to the deceleration device **30**, a control device **50** that controls the drive motor **20**, a parking mechanism **6**, and a drive unit case **11** that accommodates the above-described components.

(18) The drive unit case **11** includes a main case **111**, a left side cover **112** covering a left side surface of the main case **111**, and a right side cover **113** covering a right side surface of the main case **111**.

(19) The drive unit case **11** includes a motor chamber **12** that accommodates the drive motor **20**, a gear chamber **13** that accommodates the deceleration device **30**, a chain chamber **14** that accommodates the chain transmission mechanism **4** and the parking mechanism **6**, and a controller chamber **15** that accommodates the control device **50**. The motor chamber **12** and the gear chamber **13** are formed side by side in the front-rear direction such that the motor chamber **12** is located on the front side and the gear chamber **13** is located on the rear side. The chain chamber **14** is formed on the left side of the motor chamber **12** and the gear chamber **13** and is formed by the main case **111** and the left side cover **112**. The controller chamber **15** is formed on the right side of the motor chamber **12** and the gear chamber **13** and is formed by the main case **111** and the right side cover **113**.

(20) The drive motor **20** is a so-called inner rotor motor that includes a drive shaft **21**, a rotor **22** that is attached to the drive shaft **21** and rotates integrally with the drive shaft **21**, and a stator **23** that is disposed on a radial direction outer side of the rotor **22** in a manner of facing the rotor **22** in the radial direction with a slight gap therebetween.

(21) In the present embodiment, in the drive unit **10**, the drive motor **20** is disposed such that an axial direction (that is, the drive shaft **21**) is horizontally oriented in the left-right direction. In this way, since the drive shaft **21** is oriented in the horizontal direction, an upper-lower dimension of the drive unit **10** can be compact.

(22) The stator **23** includes a stator core **231** and a coil **232** that is attached to the stator core **231** and includes a plurality of windings of a U-phase, a V-phase, and a W-phase.

(23) A drive sprocket **21a** around which a power transmission chain **40** of the chain transmission mechanism **4** is wound is attached to a left end of the drive shaft **21**. The drive sprocket **21a** rotates integrally with the drive shaft **21**.

(24) The chain transmission mechanism **4** includes the drive sprocket **21a** attached to the drive shaft **21**, a driven sprocket **311a** attached to an input shaft **311** of a planetary gear mechanism **31** to be described later on the same plane as the drive sprocket **21a**, and the power transmission chain **40** wound around the drive sprocket **21a** and the driven sprocket **311a**. The driven sprocket **311a** has a larger diameter than the drive sprocket **21a**, and the number of teeth of the driven sprocket **311a** is larger than the number of teeth of the drive sprocket **21a**.

(25) As shown in FIGS. 3 and 4, the deceleration device **30** includes the planetary gear mechanism **31** and a differential gear mechanism **32**.

(26) The planetary gear mechanism **31** includes an input shaft **311**, a sun gear **312**, a plurality of planetary pinion shafts **313**, the same number of stepped pinions **314** as the planetary pinion shafts **313**, a planetary carrier **316**, and a ring gear **317**.

(27) In the present embodiment, the planetary gear mechanism **31** is aligned behind the drive motor **20** in the front-rear direction. The planetary gear mechanism **31** is disposed such that an axial direction (that is, the input shaft **311**) is parallel to the axial direction of the drive motor **20** and is oriented in the left-right direction. The input shaft **311** of the planetary gear mechanism **31** is disposed at substantially the same height as the drive shaft **21** of the drive motor **20** in the upper-lower direction. Further, an outer diameter dimension of the planetary gear mechanism **31** is substantially the same dimension as an outer diameter dimension of the drive motor **20**, and a height of the drive unit **10** is small in the upper-lower direction.

(28) The input shaft **311** is a hollow shaft into which a left drive shaft **325L** to be described later is inserted. The driven sprocket **311a** around which the power transmission chain **40** of the chain transmission mechanism **4** is wound is attached to a left end of the input shaft **311**. The driven sprocket **311a** rotates integrally with the input shaft **311**.

(29) The sun gear **312** is an external gear provided on the input shaft **311**, and rotates integrally with the input shaft **311** about the same rotation axis.

(30) The plurality of planetary pinion shafts **313** are disposed at equal intervals in a circumferential direction along an outer peripheral surface of the sun gear **312** in a manner of being oriented in the left-right direction parallel to the input shaft **311** on a radial direction outer side of the sun gear **312**.

(31) The stepped pinion **314** including a first planetary gear **314a** and a second planetary gear **314b** that rotate integrally is pivotally supported by each planetary pinion shaft **313** in a freely rotatable manner.

(32) Each first planetary gear **314a** is an external gear that is disposed on the outer peripheral surface of the sun gear **312** and meshes with the sun gear **312**.

(33) Each second planetary gear **314b** is an external gear that is disposed on an inner peripheral surface of the ring gear **317** and meshes with the ring gear **317**.

(34) The planetary carrier **316** connects the planetary pinion shafts **313**. The planetary carrier **316** is rotatable about a rotation axis coaxial with the input shaft **311** (and the sun gear **312**) integrally with the planetary pinion shafts **313**.

(35) Therefore, the stepped pinion **314** including the first planetary gear **314a** and the second planetary gear **314b** is freely rotatable about the planetary pinion shaft **313** as an axis, and is freely revolvable about the rotation axis coaxial with the input shaft **311** (and the sun gear **312**) integrally with the planetary pinion shaft **313**. The planetary carrier **316** rotates integrally with a revolutionary motion of the stepped pinion **314** on the rotation axis coaxial with the input shaft **311** (and the sun gear **312**).

(36) The ring gear **317** is an annular internal gear whose inner peripheral surface meshes with each second planetary gear **314b**. In the present embodiment, the ring gear **317** is fixed to the drive unit case **11**, and the ring gear **317** does not rotate.

(37) The differential gear mechanism **32** includes a differential case **321**, a differential pinion shaft **322** supported by the differential case **321**, a first bevel gear **323a** and a second bevel gear **323b** pivotally supported by the differential pinion shaft **322** in a freely pivotable manner, a left side gear **324L** and a right side gear **324R** meshing with the first bevel gear **323a** and the second bevel gear **323b**, and a left drive shaft **325L** and a right drive shaft **325R**.

(38) The differential case **321** is formed integrally with the planetary carrier **316** of the planetary gear mechanism **31**. Therefore, the differential case **321** rotates about the rotation axis coaxial with the input shaft **311** integrally with the planetary carrier **316** of the planetary gear mechanism **31**.

(39) Next, a power transmission path of power output from the drive motor **20** will be described.

(40) Power generated by the drive motor **20** is output from the drive shaft **21**, and the drive sprocket **21a** attached to the drive shaft **21** rotates integrally with the drive shaft **21**. When the drive

sprocket **21a** rotates, the driven sprocket **311a** rotates due to the power transmission chain **40** wound around the drive sprocket **21a** and the driven sprocket **311a** attached to the input shaft **311** of the planetary gear mechanism **31**. The input shaft **311** of the planetary gear mechanism **31** rotates integrally with the driven sprocket **311a**. At this time, since the number of teeth of the driven sprocket **311a** is larger than the number of teeth of the drive sprocket **21a**, the rotation of the drive shaft **21** is reduced via the drive sprocket **21a**, the power transmission chain **40**, and the driven sprocket **311a**, and is input to the input shaft **311** of the planetary gear mechanism **31**.

(41) In the planetary gear mechanism **31**, the power input to the input shaft **311** is transmitted to the stepped pinion **314** via the sun gear **312**. The stepped pinion **314** rotates while revolving. The planetary carrier **316** rotates integrally with the revolution of the stepped pinion **314**. In the planetary gear mechanism **31**, since the ring gear **317** is fixed, the rotation of the input shaft **311** is reduced at a predetermined reduction ratio and transmitted to the planetary carrier **316**.

(42) In the differential gear mechanism **32**, the differential case **321** is formed integrally with the planetary carrier **316** of the planetary gear mechanism **31**, and thus rotates integrally with the rotation of the planetary carrier **316**. Therefore, the power input to the input shaft **311** of the planetary gear mechanism **31** is reduced at the predetermined reduction ratio and input to the differential case **321** via the planetary carrier **316**.

(43) Therefore, the power output from the drive shaft **21** is input to the differential case **321** of the differential gear mechanism **32** via the power transmission chain **40** and the planetary gear mechanism **31**, and the differential pinion shaft **322** revolves about the rotation axis of the differential case **321** together with the differential case **321**.

(44) When the vehicle **V** travels straight, there is no rotation difference between the left and right rear wheels **RW**, and thus the left side gear **324L** and the right side gear **324R** that mesh with the first bevel gear **323a** and the second bevel gear **323b** rotate according to rotation of the differential pinion shaft **322**. The left rear wheel **RW** rotates when the left drive shaft **325L** rotates integrally with the rotation of the left side gear **324L**, and the right rear wheel **RW** rotates when the right drive shaft **325R** rotates integrally with the rotation of the right side gear **324R**.

(45) When the vehicle **V** turns, the differential pinion shaft **322** revolves, the first bevel gear **323a** and the second bevel gear **323b** rotate such that a rotation speed of the rear wheel **RW** on an inner side during turning decreases whereas a rotation speed of the rear wheel **RW** on an outer side during turning increases, and meanwhile, the left side gear **324L** and the right side gear **324R** that mesh with the first bevel gear **323a** and the second bevel gear **323b** rotate at different rotation speeds such that the rotation speed of the rear wheel **RW** on the inner side during turning decreases whereas the rotation speed of the rear wheel **RW** on the outer side during turning increases. The left rear wheel **RW** rotates when the left drive shaft **325L** rotates integrally with the rotation of the left side gear **324L**, and the right rear wheel **RW** rotates when the right drive shaft **325R** rotates integrally with the rotation of the right side gear **324R**. Therefore, when the vehicle **V** turns, the left drive shaft **325L** and the right drive shaft **325R** rotate such that the rotation speed of the rear wheel **RW** on the inner side during turning decreases whereas the rotation speed of the rear wheel **RW** on the outer side during turning increases.

(46) In this way, as indicated by arrows in FIG. 3, the power output from the drive motor **20** is reduced via the drive sprocket **21a**, the driven sprocket **311a**, and the power transmission chain **40**, then is input to the deceleration device **30**, is further reduced by the planetary gear mechanism **31**, and the power is appropriately distributed and transmitted to the left and right rear wheels **RW** by the differential gear mechanism **32**.

(47) Next, the chain chamber **14** and the parking mechanism **6** disposed in the chain chamber **14** will be described.

(48) As shown in FIG. 5, the chain chamber **14** is a space formed by assembling a partition wall provided at the main case **111** and a recess provided in the left side cover **112** in a manner of facing each other. In the chain chamber **14**, the drive sprocket **21a** attached to the drive shaft **21** is

disposed on the front side, the driven sprocket **311a** attached to the input shaft **311** of the planetary gear mechanism **31** is disposed on the rear side, and the power transmission chain **40** is wound around the drive sprocket **21a** and the driven sprocket **311a**.

(49) The parking mechanism **6** includes a parking gear **61**, a parking shaft **62**, a parking pole **63**, a parking rod **64**, a control shaft **65**, and a detent lever **66**.

(50) The parking gear **61** is interposed in the power transmission path in the drive unit **10**, and transitions between a locked state of being engaged with the parking pole **63** and preventing rotation, and an unlocked state of not being engaged with the parking pole **63** and allowing rotation. The parking mechanism **6** enters a parking state in which the power transmission path of the drive unit **10** is braked when the parking gear **61** is brought into the locked state, and enters a non-parking state in which the power transmission path of the drive unit **10** is not braked when the parking gear **61** is brought into the unlocked state.

(51) The parking gear **61** is disposed in a manner of being rotatable integrally with the drive shaft **21** of the drive motor **20**. Accordingly, a degree of freedom in layout of the parking gear **61** and the parking mechanism **6** can be improved, and the parking gear **61** can be disposed without increasing a size of the drive unit **10** even when there is no space for disposing the parking gear **61** downstream of the drive shaft **21** of the drive motor **20**.

(52) The parking gear **61** is disposed at a position different from that of the drive sprocket **21a** in a rotation axis direction of the drive shaft **21** of the drive motor **20**. Specifically, at the drive shaft **21** of the drive motor **20**, the parking gear **61** is disposed on a tip end side (left side) of the drive shaft **21** relative to the drive sprocket **21a**. Accordingly, interference between the parking gear **61** and the drive sprocket **21a** and interference between the parking gear **61** and the power transmission chain **40** can be prevented.

(53) The parking shaft **62** is supported by the drive unit case **11** and has a rotation axis parallel to the rotation axis of the drive shaft **21** of the drive motor **20**.

(54) The parking pole **63** is supported in a manner of being swingable about the parking shaft **62**, and swings on the same plane as the parking gear **61**. The parking pole **63** includes an engagement claw portion **63a** at a tip end thereof, and transitions between an engaged state of being engaged with an engagement groove portion **61b** of the parking gear **61** and a disengaged state of not being engaged with the engagement groove portion **61b** of the parking gear **61**. A pole spring **67** is disposed between the parking pole **63** and the drive unit case **11**, and the parking pole **63** is biased by the pole spring **67** to be in the disengaged state.

(55) The parking rod **64** is supported by the drive unit case **11** via a parking rod holder **68** in a manner of being movable forward and backward, and moves forward and backward on the same plane as the parking gear **61**. The parking rod **64** moves forward and backward between an advance position where the parking pole **63** is pressed such that the parking pole **63** is brought into the engaged state and a retreat position where the parking rod **64** is separated from the parking pole **63** such that the parking pole **63** is brought into the disengaged state. A rod spring **69** is disposed between the parking rod **64** and the parking rod holder **68**, and the parking rod **64** is biased by the rod spring **69** so as to be located at the retreat position.

(56) The control shaft **65** has a rotation axis parallel to the rotation axis of the drive shaft **21** of the drive motor **20**, and is rotatably supported by the drive unit case **11** via a support boss **70**. The control shaft **65** is connected to a parking operation tool (not shown) and rotates in response to an operation on the parking operation tool. An arrangement of the control shaft **65** will be described later.

(57) The detent lever **66** is supported to be integrally rotatable with the control shaft **65** and swings on the same plane as the parking gear **61**. The detent lever **66** swings between a first position where the parking rod **64** is moved to the advance position and a second position where the parking rod **64** is moved to the retreat position according to rotation of the control shaft **65**. Accordingly, the parking mechanism **6** can be switched between the parking state and the non-parking state

according to an operation on the parking operation tool.

(58) The detent lever **66** includes a first extending portion **661** that extends in an orthogonal direction from a rotation axis and is connected to the parking rod **64**, and a second extending portion **662** that extends from the rotation axis in a direction opposite to the first extending portion **661** and engages with a detent spring **72**. The second extending portion **662** has at least two detent grooves **662a** and **662b** arranged in a swinging direction of the detent lever **66**, the detent lever **66** is held at the first position by engagement between the first detent groove **662a** and the detent spring **72**, and the detent lever **66** is held at the second position by engagement between the second detent groove **662b** and the detent spring **72**.

(59) Next, the arrangement of the control shaft **65** will be further described with reference to FIGS. **6** to **10**.

(60) As schematically shown in FIG. **6**, the control shaft **65** is disposed in the power transmission chain **40** when viewed in the rotation axis direction of the drive shaft **21** of the drive motor **20**. That is, the control shaft **65** is disposed in a region surrounded by the power transmission chain **40**. In this way, the control shaft **65** can be disposed in the power transmission chain **40** which tends to be a dead space, and the drive unit **10** can be downsized.

(61) However, in the power transmission chain **40**, in order to perform appropriate power transmission, a clearance that allows appropriate swinging between the sprockets **21a** and **311a** is set. Therefore, when the control shaft **65** is disposed in the power transmission chain **40**, the swinging power transmission chain **40** may come into contact with the support boss **70** that supports the control shaft **65**. Although it is conceivable to dispose the support boss **70** at a shifted position where the power transmission chain **40** is not interfered with in the rotation axis direction (left-right direction), in order to reduce the size of the drive unit **10**, as shown in FIG. **7**, there may be a case where the support boss **70** necessarily overlaps the power transmission chain **40** when viewed from above at least on one side (the right side in the present embodiment).

(62) In the drive unit **10** of the present embodiment, an arrangement condition of the control shaft **65** (support boss **70**) is set in order to prevent the power transmission chain **40** that swings when the control shaft **65** is disposed in the power transmission chain **40** from coming into contact with the support boss **70** that supports the control shaft **65**. First, an example of the arrangement condition will be described with reference to FIG. **8**, and another example of the arrangement condition will be described with reference to FIGS. **9** and **10**. Broken lines in FIGS. **8** and **9** show a state in which an upper part of the power transmission chain **40** swings to an innermost side.

(63) An example of the arrangement condition includes a first condition and a second condition as conditions to be simultaneously satisfied.

(64) As shown in FIG. **8**, the first condition is that $D < X - (d1 + d2)/2$, where D is a diameter of the support boss **70** of the control shaft **65**, X is an axial distance between the drive sprocket **21a** (the drive shaft **21** of the drive motor **20**) and the driven sprocket **311a** (the input shaft **311** of the planetary gear mechanism **31**), $d1$ is a diameter of the drive sprocket **21a**, and $d2$ is a diameter of the driven sprocket **311a**.

(65) By satisfying the first condition, the support boss **70** can be disposed between the shafts of the drive sprocket **21a** and the driven sprocket **311a** without interfering with the drive sprocket **21a** and the driven sprocket **311a**. The axial distance refers to a distance between axial centers, and the diameters of the sprockets **21a** and **311a** each refer to a distance to an outermost peripheral surface passing through each axial center.

(66) As shown in FIG. **8**, the second condition is that a rotation axis O of the control shaft **65** is disposed in a region at a distance longer than $D/2 + A$ from a circumferential line $L1$ connecting a chain track innermost diameter circle $21ap$ (a circle having a diameter $dp1$) of the drive sprocket **21a** and a chain track innermost diameter circle $311ap$ (a circle having a diameter $dp2$) of the driven sprocket **311a**, where A is a swing amplitude of the power transmission chain **40**. The chain track innermost diameter circles $21ap$ and $311ap$ are lines (circles) connecting innermost diameter

portions of the power transmission chain **40** engaged with the sprockets **21a** and **311a**.

(67) By satisfying the second condition, it is possible to avoid contact of the power transmission chain **40** with the support boss **70** even when the power transmission chain **40** swings and passes as far inward as possible.

(68) On the other hand, another example of the arrangement condition includes the above-described first condition and a third condition different from the above-described second condition as conditions to be simultaneously satisfied.

(69) As shown in FIG. **9**, the third condition is that the rotation axis O of the control shaft **65** is disposed in a region at a distance longer than D/2 from an arc L**3** that is tangent to three lines, that is, a parallel line L**2** that is A away on the support boss **70** side from the circumferential line L**1** common to the chain track innermost diameter circle **21ap** of the drive sprocket **21a** and the chain track innermost diameter circle **311ap** of the driven sprocket **311a**, the chain track innermost diameter circle **21ap** of the drive sprocket **21a**, and the chain track innermost diameter circle **311ap** of the driven sprocket **311a**, where A is the swing amplitude of the power transmission chain **40**. The arc L**3** is substantially equal to a trajectory of the power transmission chain **40** that swings to the innermost side.

(70) A radius r of the arc L**3** can be calculated using the following formula. As shown in FIG. **10**, the following formula (1) is valid, where B is a distance between contact points of the circumferential line L**1** common to the chain track innermost diameter circle **21ap** of the drive sprocket **21a** and the chain track innermost diameter circle **311ap** of the driven sprocket **311a**.

$$r.\sup.2=(r-A).\sup.2+(B/2).\sup.2 \quad (1)$$

(71) When this formula is expanded, the following formula (2) for obtaining the radius r of the arc L**3** based on the swing amplitude A and the distance B between the contact points of the circumferential line L**1** is obtained. The swing amplitude A and the distance B between the contact points of the circumferential line L**1** are values determined by the diameters of the drive sprocket **21a** and the driven sprocket **311a**, the axial distance between the drive sprocket **21a** and the driven sprocket **311a**, and a length and properties of the power transmission chain **40**.

$$r=(A/2)+(B.\sup.2/8A) \quad (2)$$

(72) By satisfying the third condition, it is possible to avoid contact of the power transmission chain **40** with the support boss **70** even when the power transmission chain **40** swings and passes as far inward as possible.

(73) Although the embodiment of the present invention has been described above with reference to the accompanying drawings, it is needless to say that the present invention is not limited to the embodiment. It is apparent that those skilled in the art can conceive of various modifications and alterations within the scope described in the claims, and it is understood that such modifications and alterations naturally fall within the technical scope of the present invention. In addition, the constituent elements in the above embodiment may be freely combined without departing from the gist of the invention.

(74) For example, the power transmission chain **40** is shown as an example of the endless flexible member, but the endless flexible member is not limited thereto, and may be a belt.

(75) In this specification, at least the following matters are described. Although corresponding constituent elements or the like in the embodiment described above are shown in parentheses, the present invention is not limited thereto.

(76) (1) A power transmission device (drive unit **10**) including: a drive source (drive motor **20**); a transmission portion (chain transmission mechanism **4**) configured to convert power of the drive source and transmit the power to a driven portion; and a case (drive unit case **11**) accommodating the drive source and the transmission portion, in which the power transmission device further includes a drive sprocket (drive sprocket **21a**) disposed at an output shaft (drive shaft **21**) of the drive source, a driven sprocket (driven sprocket **311a**) disposed on the same plane as the drive sprocket, an endless flexible member (power transmission chain **40**) wound around the drive

sprocket and the driven sprocket, and a parking gear (parking gear **61**) disposed at the output shaft of the drive source.

(77) According to (1), a degree of freedom in layout of the parking gear is improved.

(78) (2) The power transmission device according to (1), in which the parking gear is disposed at a position different from that of the drive sprocket in a rotation axis direction of the output shaft of the drive source.

(79) According to (2), interference with the drive sprocket can be prevented.

(80) (3) The power transmission device according to (1), further including: a parking shaft (parking shaft **62**) having a rotation axis parallel to a rotation axis of the output shaft of the drive source; and a parking pole (parking pole **63**) supported by the parking shaft and swingable on a same plane as the parking gear, in which the parking pole is configured to transition between an engaged state of being engaged with the parking gear and a disengaged state of not being engaged with the parking gear, and is biased to be in the disengaged state.

(81) According to (3), the driven portion can be maintained in a parking state by engaging the parking pole with the parking gear.

(82) (4) The power transmission device according to (3), further including: a parking rod (parking rod **64**) movable forward and backward on the same plane as the parking gear, in which the parking rod is configured to move between an advance position where the parking rod is capable of pressing the parking pole and a retreat position away from the parking pole, and is biased to be located at the retreat position.

(83) According to (4), the position of the parking pole can be controlled by the parking rod.

(84) (5) The power transmission device according to (4), further including: a control shaft (control shaft **65**) having a rotation axis parallel to the rotation axis of the output shaft of the drive source; and a detent lever (detent lever **66**) supported by the control shaft and swingable on the same plane as the parking gear, in which the detent lever is configured to swing between a first position getting the parking rod moved to the advance position and a second position getting the parking rod moved to the retreat position, and is biased to be located at the second position.

(85) According to (5), the position of the parking rod can be controlled by the detent lever.

(86) (6) The power transmission device according to (5), in which the control shaft is disposed inside the endless flexible member when viewed in the rotation axis direction of the output shaft of the drive source.

(87) According to (6), since the control shaft is disposed in the endless flexible member which tends to be a dead space, a drive device can be downsized.

(88) (7) The power transmission device according to (6), further including: a support boss (support boss **70**) provided at the case and supporting the control shaft, in which a relationship of $D < X - (d_1 + d_2)/2$ is satisfied, where D is a diameter of the support boss of the control shaft, X is a distance between the output shaft of the drive source and a rotation shaft (input shaft **311**) of the driven sprocket, d_1 is a diameter of the drive sprocket, and d_2 is a diameter of the driven sprocket, and a rotation axis (rotation axis O) of the control shaft is disposed in a region at a distance longer than $D/2 + A$ from an external tangential line (circumferential line **L1**) common to a chain track innermost diameter circle (chain track innermost diameter circle **21ap**) of the drive sprocket and a chain track innermost diameter circle (chain track innermost diameter circle **311ap**) of the driven sprocket, where A is a swing amplitude of the endless flexible member.

(89) According to (7), even when the control shaft is disposed in the endless flexible member, the endless flexible member can be prevented from coming into contact with the support boss that supports the control shaft.

(90) (8) The power transmission device according to (6), further including: a support boss (support boss **70**) provided at the case and configured to support the control shaft, in which a relationship of $D < X - (d_1 + d_2)/2$ is satisfied, where D is a diameter of the support boss of the control shaft, X is a distance between the output shaft of the drive source and a rotation shaft (input shaft **311**) of the

driven sprocket, $d1$ is a diameter of the drive sprocket, and $d2$ is a diameter of the driven sprocket, and a rotation axis (rotation axis O) of the control shaft is disposed in a region at a distance longer than $D/2$ from an arc (arc L3) that is tangent to three lines: a parallel line (parallel line L2) that is away by A on a support boss side from an external tangential line common to a chain track innermost diameter circle (chain track innermost diameter circle 21ap) of the drive sprocket and a chain track innermost diameter circle (chain track innermost diameter circle 311ap) of the driven sprocket; the chain track innermost diameter circle of the drive sprocket; and the chain track innermost diameter circle of the driven sprocket, where A is a swing amplitude of the endless flexible member.

(91) According to (8), even when the control shaft is disposed in the endless flexible member, the endless flexible member can be prevented from coming into contact with the support boss that supports the control shaft.

Claims

1. A power transmission device comprising: a drive source; a transmission portion configured to convert power of the drive source and transmit the power to a driven portion; and a case accommodating the drive source and the transmission portion, wherein the transmission portion includes a drive sprocket disposed at an output shaft of the drive source, a driven sprocket disposed on a same plane as the drive sprocket, and an endless flexible member wound around the drive sprocket and the driven sprocket, the power transmission device further comprises: a parking gear disposed at the output shaft of the drive source; a parking shaft having a rotation axis parallel to a rotation axis of the output shaft of the drive source; a parking pole supported by the parking shaft and swingable on a same plane as the parking gear; a parking rod movable forward and backward on the same plane as the parking gear; a control shaft having a rotation axis parallel to the rotation axis of the output shaft of the drive source; and a detent lever supported by the control shaft and swingable on the same plane as the parking gear, the parking pole is configured to transition between an engaged state of being engaged with the parking gear and a disengaged state of not being engaged with the parking gear, and is biased to be in the disengaged state, the parking rod is configured to move between an advance position where the parking rod is capable of pressing the parking pole and a retreat position away from the parking pole, and is biased to be located at the retreat position, the detent lever is configured to swing between a first position getting the parking rod moved to the advance position and a second position getting the parking rod moved to the retreat position, and is biased to be located at the second position, and the control shaft is disposed inside the endless flexible member when viewed in the rotation axis direction of the output shaft of the drive source.

2. The power transmission device according to claim 1, wherein the parking gear is disposed at a position different from that of the drive sprocket in a rotation axis direction of the output shaft of the drive source.

3. The power transmission device according to claim 1, further comprising: a support boss provided at the case and supporting the control shaft, wherein a relationship of $D < X - (d1 + d2)/2$ is satisfied, where D is a diameter of the support boss of the control shaft, X is a distance between the output shaft of the drive source and a rotation shaft of the driven sprocket, $d1$ is a diameter of the drive sprocket, and $d2$ is a diameter of the driven sprocket, and a rotation axis of the control shaft is disposed in a region at a distance longer than $D/2 + A$ from an external tangential line common to a chain track innermost diameter circle of the drive sprocket and a chain track innermost diameter circle of the driven sprocket, where A is a swing amplitude of the endless flexible member.

4. The power transmission device according to claim 1, further comprising: a support boss provided at the case and configured to support the control shaft, wherein a relationship of $D < X - (d1 + d2)/2$ is satisfied, where D is a diameter of the support boss of the control shaft, X is a distance between the

output shaft of the drive source and a rotation shaft of the driven sprocket, d_1 is a diameter of the drive sprocket, and d_2 is a diameter of the driven sprocket, and a rotation axis of the control shaft is disposed in a region at a distance longer than $D/2$ from an arc that is tangent to three lines: a parallel line that is away by A on a support boss side from an external tangential line common to a chain track innermost diameter circle of the drive sprocket and a chain track innermost diameter circle of the driven sprocket; the chain track innermost diameter circle of the drive sprocket; and the chain track innermost diameter circle of the driven sprocket, where A is a swing amplitude of the endless flexible member.
